

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design a system for the client, DriverPass, that helps students prepare for their driving tests more effectively
- DriverPass is a company created by Liam to solve the issue of people failing DMV driving tests
- The system should provide both online learning and real driving practice in one platform
- The system must allow students to take online practice exams, attend classes, and schedule driving lessons
- It should be accessible from any computer or mobile device using the internet
- The system should also support different users such as:
 - Students (customers)
 - Instructors (drivers)
 - Secretary (scheduling and support)
 - IT officer (system management)
 - Owner (full access and reporting)
- Overall, the goal is to create a user-friendly, secure, and flexible system that improves student success rates

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass identified a real-world problem where many people fail their DMV driving tests due to lack of preparation
- Current training options are not effective or do not provide enough practice opportunities

- DriverPass wants to solve this by offering:
 - Online classes and practice tests
 - Real driving lessons with instructors
- The system must support:
 - Online access to training materials and practice exams
 - Scheduling system for driving lessons (2-hour sessions)
- Tracking of reservations (who created, updated, or canceled)
- Multiple training packages (6, 8, or 12 hours of driving + additional features)
- The system should include:
 - User account management with secure login and password reset
 - Role-based access control (different permissions for each user)
- Data tracking and reporting features
- Integration with DMV updates for current rules and test questions
- The system will be **cloud-based**, so DriverPass does not need to manage backups or security manually
- It should also include interface features like:
 - Student information forms
 - Test progress tracking (scores, status, time)
 - Driver notes and lesson details (as shown in the design sketch on page 4)

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system should allow users to:
 - Create accounts and log in securely
 - Reset passwords automatically if forgotten
- The system should provide:
 - Online practice tests with updated DMV content
 - Tracking of test progress (completed, in progress, passed, failed)
- The system should allow students to:
 - Schedule, modify, and cancel driving lessons online
 - Select available time slots and instructors
- The system should allow staff to:
 - Schedule appointments for customers (secretary role)
 - Manage student information and reservations
- The system should track:
 - Reservation details (who created/modified/canceled)
 - Lesson schedules, drivers, and assigned vehicles
- The system should support:
 - Multiple training packages and allow enabling/disabling packages
- The system should include:
 - Secure payment data handling (credit card info during registration)
- The system should generate:
 - Reports that can be downloaded (e.g., Excel reports)

- The system should:
 - Be accessible online anytime
 - Work across devices (mobile and desktop)
 - Store data securely in the cloud
- The system should:
 - Receive updates from the DMV when rules or test questions change
 - Notify the company when updates occur

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- [Insert text]

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- [Insert text]

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- [Insert text]

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- [Insert text]

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- [Insert text]

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- [Insert text]

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- [Insert text]

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- [Insert text]

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- [Insert text]

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

[Insert chart]